

RF Admittance level switch For Solid Application, Model: RFLS-300S



PRODUCT DESCRIPTION:

The RFLS-300S RF Admittance type level switch is a simple, low cost, rugged point Level Switch designed for Level detection of Bulk materials in a bins or silo. This level switch can be used for detecting high, low level or for differential level detection and is used in various level detection applications like. cement industries, material handling plants, chemical industry, power utilities industry, food industry, pulp/paper industry, pharmaceuticals industry, water/ treatment plants, foundry industry etc.

The specialty of the probe is that it tends to ignore the build-up on the probe, thus making it suitable for application, which tends to stick on the probe.

The unit has integral electronics along with time delay feature, LED indication, fail safe mode etc.

SALIENT FEATURES:

- Ø Advanced RF Admittance Technology.
- Ø Wide range of industrial applications.
- Ø Suitable for High, Low and Differential Level Detection.
- Ø Integral and Remote Electronics Type available.
- Ø Adjustable Sensitivity for reliable level detection of different materials.
- Ø Easy installation and maintenance.
- Ø Weatherproof / Flameproof enclosure as per requirement.
- Ø PTFE insulated electrode capable of withstanding temperatures up to 180°C.
- Ø No moving parts – no mechanical wear and tear.
- Ø Adjustable differential time delay with ON/OFF timing.
- Ø Field-configurable Fail-Safe High / Low selection.
- Ø Ignores build-up of sticky and conductive materials between the probe and vessel wall.
- Ø LED indication for easy status monitoring.
- Ø Suitable for applications with material build-up on the probe.
- Ø Robust construction for demanding industrial applications.
- Ø Low maintenance and long service life.

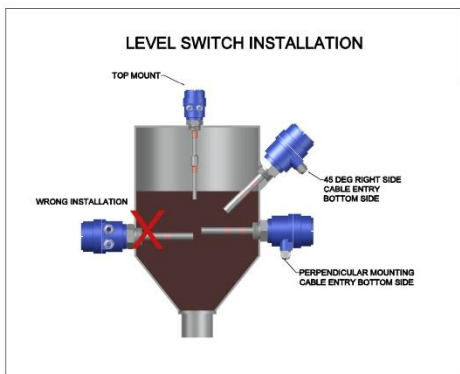


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OPERATING PRINCIPLE:

The RFLS-300S operates on the RF Admittance Principle. The probe and vessel wall form a sensing system that generates an RF electric field around the sensing probe. When the material level reaches the probe, the dielectric properties of the material cause a change in RF Admittance. this change is detected by the electronic circuit and converted into a reliable level switching output. Adjustable Sensitivity allows the instrument to be set according to different material characteristics and process conditions. the Build-up Rejection feature minimizes the effect of sticky or conductive material coating on the probe, ensuring reliable level detection even in applications where material build-up occurs. the instrument can be configured for High Level, Low Level and Differential Level Detection. Adjustable ON/OFF Time Delay and Fail-Safe High/Low Selection are available as required. the level detection signal is provided through Relay / PNP Output. With no moving parts, the instrument offers low maintenance, reduced mechanical wear and long service life.

PROBE CONSTRUCTION – COAT Guard FOR MATERIAL BUILD-UP:



A normal probe may tend to malfunction when material builds up on the probe. The COAT Guard electrode is designed with special Build-up Sensing Technology to minimize the effect of material coating on the probe. The COAT Guard is electrically insulated from the active sensing probe and is driven with the same RF frequency and polarity as the active probe. When material builds up on the probe, the RF current from

the COAT Guard tends to saturate the build-up area between the COAT Guard and the vessel wall. This significantly reduces the effect of the build-up on the active sensing probe and helps prevent false level switching. When the actual material level rises and reaches the sensing probe, the RF current flows from the active sensing probe to ground through the material, producing a reliable level detection signal. The COAT Guard Technology provides stable and reliable operation in applications involving sticky, conductive and coating-forming materials. It helps maintain consistent level detection even when material build-up occurs on the probe.

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COMPONENTS OF LEVEL SWITCH EXPLAINED IN FIGURE

Integral Type

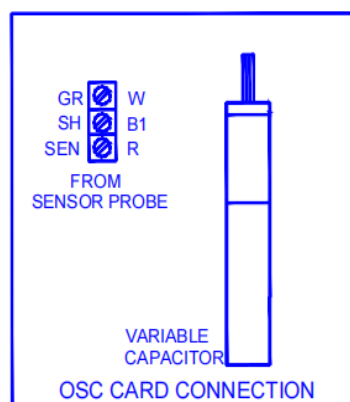
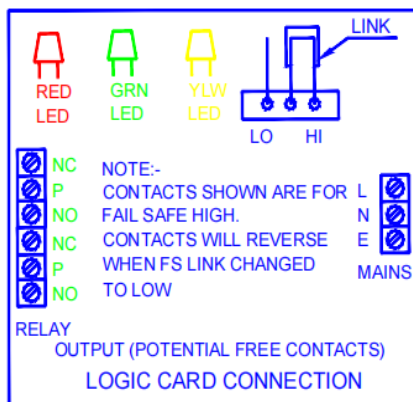


Remote Type



Controller / RF Cable / Remote Probe

ELECTRICAL CONNECTION ARE UNIVERSAL /230 VAC/110VAC/24 VDC



DATA SHEET

RFLS-300– RF Admittance Level Switch

Selection / Ordering Information

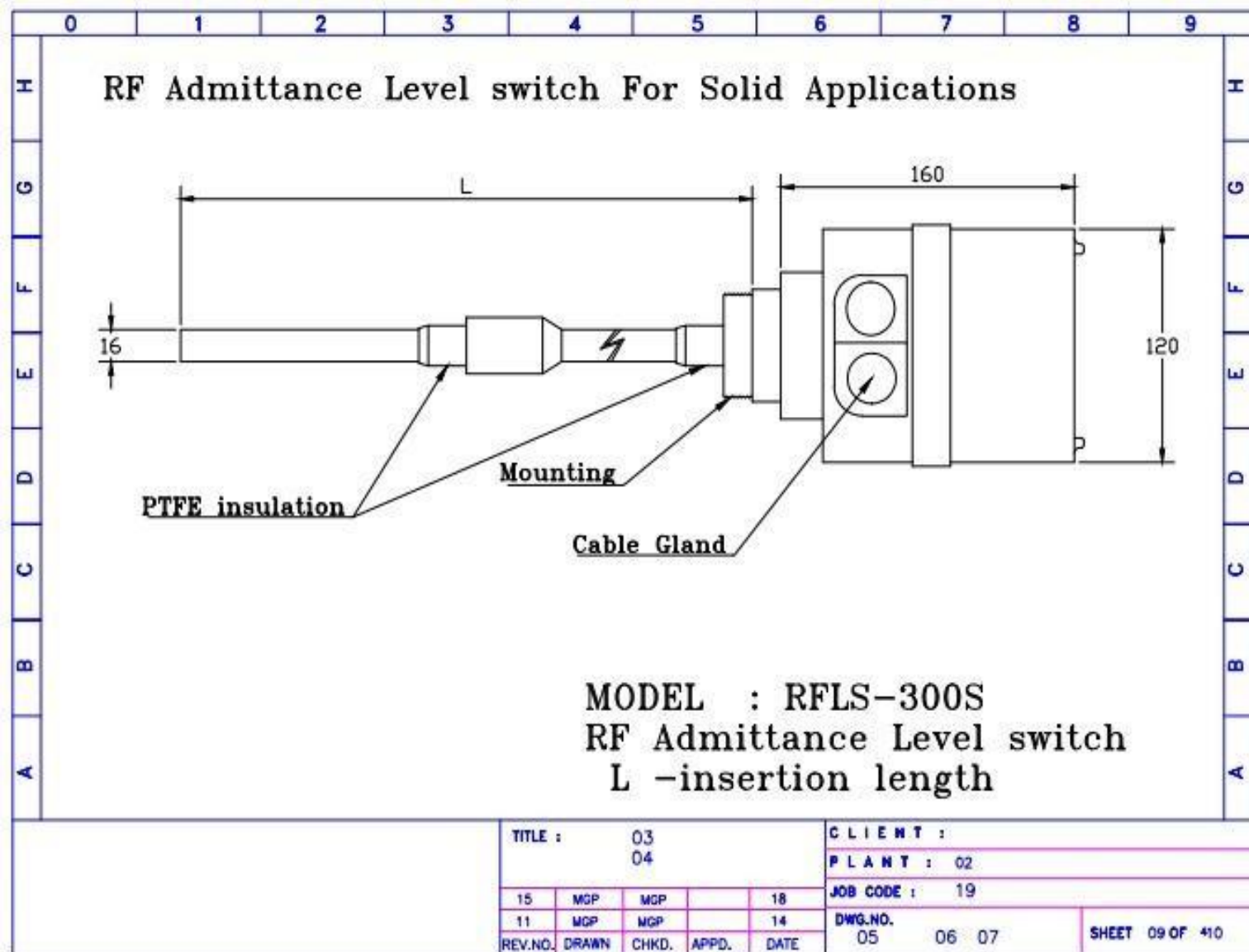
Selection	Code	Description
Model No.	RFLS-300	RF Admittance Level Switch
Application	S	Solids
Mounting Type	IN/RE	Integral / Remote
Probe Type	RD / RP	Rigid Rod / Rope Flexible
Insulation	FP/PT	Full PTFE / Part PTFE
Probe Diameter	D1 / D2	Standard 16mm / Heavy Duty 30mm
Probe Material	04 /4L/ 16/ 6L/MS/ SM	SS304/SS304L / SS316/ SS316L /Mild steel/ Special material
Power Supply	AC/DC/A1/UN	230VAC / 24VDC / 110VAC / Universal
Output	DP/SP/PN	DPDT Relay /SPDT Relay / PNP Relay
Switching Delay	ND / DA	No Delay / Adjustable
Indication	L	LED

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<u>Enclosure</u>	WP / FL	Weatherproof / Flameproof
<u>Connection</u>	T	Threaded
	F	Flanged
<u>Mounting</u>	B1 / B2 / B3/B4	1" / 1½" / 2" BSP/ Special Mounting
	N1 / N2 / N3/N4	1" / 1½" / 2" NPT/ Special Mounting
<u>Flange Rating</u>	A1 / A3 / A6	ANSI 150# / ANSI 300# /ANSI 600#
<u>Flange Facing</u>	RF / FF	Raised Face / Flat Face
<u>MOC-Process Connection</u>	S4 / S6/ S16/MS/SL	SS304 / SS316 /SS316L /Mild steel / Special material
<u>Flange Size</u>	01 / 15 / 02 / 03 / 04/PS	1" / 1½" / 2" / 3" / 4"/ Special Process
<u>Pressure</u>	P1 /P2/P3/P4	Standard 0 to 2KG/ 2 To 10kg /10kg to 40kg / Customised
<u>Temperature</u>	T1/ T2/TS	Max -20-150°C / 200°C / Customer Specified Special Designed
<u>High Temperature stand off</u>	H8/H1/H2/H3/HS	80MM/150 MM /200MM /300 MM /Special standoff Extension
<u>Remote Cable</u>	05 / 10 / 15 / 20 / 25 / 30/SR	5 / 10 / 15 / 20 / 25 / 30 m/ Special Customer Requirement
<u>Cable Gland</u>	G1/ G2 / G3/ G4	PG13.5 / ½"NPT / ¾" ET/ PG16
<u>Ground / Reference</u>	GS / GX	Standard / Special
<u>Material Certificate</u>	M0 / MC	Not Required / Material test certificate
<u>Test Certificate</u>	TC / CC	Test Certificate / Calibration Certificate
<u>Documentation</u>	DS / GA / IR	Datasheet / GA / Inspection Report
<u>Tagging</u>	TAG	Customer Tag No.
<u>Probe Length</u>	XXXX	200 mm to customised

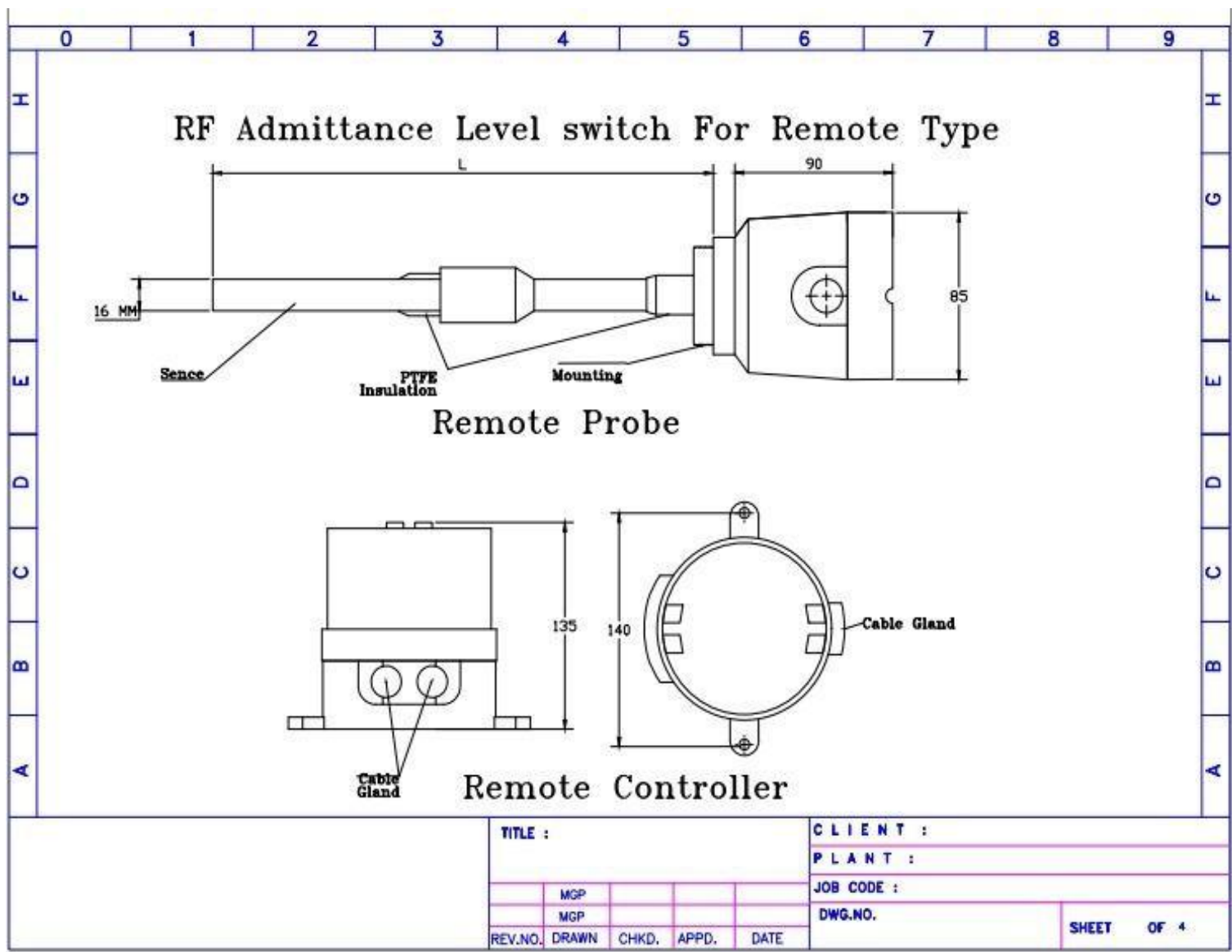
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Mechanical Drawing: Integral Type:



RF Admittance level switch For Solid Application, Model: RFLS-300S

Mechanical Drawing: Remote Type:



Manufactured By:

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